

Qualification Pack



Junior Parataxonomist

QP Code: SGJ/Q0701

Version: 1.0

NSQF Level: 4.5

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SGJ/Q0701: Junior Parataxonomist

Brief Job Description

The Junior Parataxonomist supports biodiversity conservation through field-based identification, classification, and documentation of flora and fauna. They contribute to People's Biodiversity Registers, implement the Biological Diversity Act, 2002, support Biodiversity Management Committees, and use GIS tools for conservation planning. With knowledge of national and international biodiversity frameworks, they connect scientific data with local conservation efforts to ensure sustainable bioresource management.

Personal Attributes

The supports biodiversity conservation through field-based identification, classification, and documentation of flora and fauna. They contribute to People's Biodiversity Registers, implement the Biological Diversity Act, 2002, support Biodiversity Management Committees, and use GIS tools for conservation planning. With knowledge of national and international biodiversity frameworks, they connect scientific data with local conservation efforts to ensure sustainable bioresource management.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0703: Explore Floristic Diversity, Endemism, and Economic Botany for Conservation](#)
2. [SGJ/N7001: Conduct Field Surveys, Collect, Preserve, and Identify Flora](#)
3. [SGJ/N7002: Understand Biodiversity in National and International Contexts](#)
4. [SGJ/N7003: Study and Classify Invertebrates and Vertebrates Using Taxonomy](#)
5. [SGJ/N7004: Perform Faunal Surveys, Assess Biodiversity, and Apply Conservation Practices](#)
6. [SGJ/N7005: Implement the Biological Diversity Act, 2002 and Institutional Mechanisms](#)
7. [SGJ/N7006: Constitute Biodiversity Management Committees and Operate them Effectively](#)
8. [SGJ/N7007: Document People Biodiversity Registers and Develop Bioresource Strategies](#)
9. [SGJ/N7008: Introduce Remote Sensing and GIS Techniques for Conservation Efforts](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	Green Jobs
Sub-Sector	Forestry
Occupation	Conservationist
Country	India
NSQF Level	4.5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2131
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) (in Field related with Biology/ life science) with NA of experience OR 12th grade Pass (in Field related with Biology/ life science) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (4) with 1.5 years of experience in Field related with Biology/ life science OR Previous relevant Qualification of NSQF Level (3.5) with 3 Years of experience in Field related with Biology/ life science
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-4.5-ES-04261-2025
NQR Version	1

Remarks:



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Total 480 hours: (Theory: 245 hours, Practical: 155 hours, OJT: 80 hours)

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SGJ/N0703: Explore Floristic Diversity, Endemism, and Economic Botany for Conservation

Description

Endemism, and Economic Botany for Conservation

Scope

The scope covers the following :

- Explore Floristic Diversity, Endemism, and Economic Botany for Conservation

Elements and Performance Criteria

Floristic diversity and Endemism. Economic Botany and Traditional Knowledge and Wildlife (Flora) Protection and Conservation

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss about the Major ecosystems- To observe and analyze different ecosystems and their components in their natural settings
- PC2.** Discuss the Vegetation types of India, Phytogeographical regions of India- To gain knowledge of various plant groups and phytogeographical regions in India, Biodiversity Hotspots in India and Endemism in Indian Flora with Special Emphasis on Flowering Plants
- PC3.** Discuss about the useful Plants and Plant Products; Medicinal Plants and Folklore; Folklore and Traditional Indian Medicinal Systems; Indigenous Tribal Communities and Traditional Knowledge (Region-Wise)
- PC4.** Learn about the flora and conservation practices associated with sacred groves.
- PC5.** Overview of IUCN Red List
- PC6.** Overview of CITES
- PC7.** Discuss about CITES Implementation
- PC8.** Explain about Non-Detriment Finding (NDF) for Plant Species
- PC9.** Explain about the Identification of Illegally Traded Plant Species Using Pharmacognosy Techniques
- PC10.** Discuss about in situ/ex situ conservation
- PC11.** Understand the economic significance of plants and their products.
- PC12.** Identify key plants used in various industries and their roles
- PC13.** Understand the principles of traditional Indian medicinal systems and their applications.
- PC14.** Documentation of traditional plant knowledge and practices among indigenous communities.
- PC15.** Understand the ecological and cultural importance of sacred groves.
- PC16.** Understand the categories and criteria used to assess the conservation status of plant species.
- PC17.** Identify threatened flora in specific regions and their conservation needs.
- PC18.** Recognize CITES-listed plant species and understand the impact of international trade on their conservation.

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PC19. Develop skills in identifying and authenticating plant species using scientific techniques.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about the Major ecosystems- To observe and analyze different ecosystems and their components in their natural settings
- KU2.** Learn uses of medicinal plants and traditional knowledge of tribal communities.
- KU3.** Recognize the cultural and ecological value of sacred groves.
- KU4.** Understand IUCN Red List, CITES, and NDF for plant conservation.
- KU5.** Identify economically important plants and their industrial applications.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** bserve and analyze different ecosystems and their components in their natural settings
- GS2.** Document traditional knowledge of medicinal plants and folklore practices among indigenous communities.
- GS3.** Apply knowledge of conservation principles, including IUCN, CITES, and NDF, in plant protection efforts.
- GS4.** Communicate effectively with local communities to understand and record ethnobotanical practices.
- GS5.** Evaluate the economic and ecological importance of plants used in industries and traditional medicine.
- GS6.** Promote awareness and practices related to the conservation of threatened flora and sacred groves.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Floristic diversity and Endemism. Economic Botany and Traditional Knowledge and Wildlife (Flora) Protection and Conservation</i>	53	47	-	-
PC1. Discuss about the Major ecosystems- To observe and analyze different ecosystems and their components in their natural settings	6	-	-	-
PC2. Discuss the Vegetation types of India, Phytogeographical regions of India- To gain knowledge of various plant groups and phytogeographical regions in India, Biodiversity Hotspots in India and Endemism in Indian Flora with Special Emphasis on Flowering Plants	7	-	-	-
PC3. Discuss about the useful Plants and Plant Products; Medicinal Plants and Folklore; Folklore and Traditional Indian Medicinal Systems; Indigenous Tribal Communities and Traditional Knowledge (Region-Wise)	5	-	-	-
PC4. Learn about the flora and conservation practices associated with sacred groves.	6	-	-	-
PC5. Overview of IUCN Red List	9	-	-	-
PC6. Overview of CITES	5	-	-	-
PC7. Discuss about CITES Implementation	5	-	-	-
PC8. Explain about Non-Detriment Finding (NDF) for Plant Species	5	-	-	-
PC9. Explain about the Identification of Illegally Traded Plant Species Using Pharmacognosy Techniques	3	-	-	-
PC10. Discuss about in situ/ex situ conservation	2	-	-	-
PC11. Understand the economic significance of plants and their products.	-	6	-	-
PC12. Identify key plants used in various industries and their roles	-	7	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Understand the principles of traditional Indian medicinal systems and their applications.	-	4	-	-
PC14. Documentation of traditional plant knowledge and practices among indigenous communities.	-	6	-	-
PC15. Understand the ecological and cultural importance of sacred groves.	-	6	-	-
PC16. Understand the categories and criteria used to assess the conservation status of plant species.	-	6	-	-
PC17. Identify threatened flora in specific regions and their conservation needs.	-	5	-	-
PC18. Recognize CITES-listed plant species and understand the impact of international trade on their conservation.	-	5	-	-
PC19. Develop skills in identifying and authenticating plant species using scientific techniques.	-	2	-	-
NOS Total	53	47	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0703
NOS Name	Explore Floristic Diversity, Endemism, and Economic Botany for Conservation
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N7001: Conduct Field Surveys, Collect, Preserve, and Identify Flora

Description

This NOS provides step-by-step techniques for field collection, proper specimen handling, drying, and long-term preservation for botanical research.

Scope

The scope covers the following :

- Learn practical methods for collecting, pressing, mounting, and preserving plant specimens.

Elements and Performance Criteria

Field surveys, collections, preservation and identification of flora

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss about the morphological characters of Flowering Plants
- PC2.** Explain the identification Techniques Method of Dissecting Floral Parts and Observation
- PC3.** Explain the identification Techniques of local plant species using field guides and identification keys
- PC4.** Explain to understand and use scientific names
- PC5.** Discuss about Collection of Plants and Field Data; Method of Pressing, Mounting, and Preservation of Plant Specimens
- PC6.** Explain the procedure of Incorporation and Curation of Specimens (Herbarium/ Museum)
- PC7.** Develop a deep understanding of plant morphology and terminology.
- PC8.** Gain hands-on experience in dissecting, collecting, preserving, and curating plant specimens
- PC9.** Enhance skills in identifying plants using various botanical resources and keys.
- PC10.** Build practical experience in field data Collection and specimen documentation.
- PC11.** Understand best practices in plant preservation methods

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the external features of flowering plants, including root, stem, leaf, inflorescence, flower, fruit, and seed structures.
- KU2.** Knowledge of dissecting floral parts and observing reproductive structures to identify plant families and species.
- KU3.** Familiarity with using field guides, dichotomous keys, floras, and botanical resources for accurate species identification.
- KU4.** Understanding the principles of binomial nomenclature, taxonomic hierarchy, and correct usage of scientific names.
- KU5.** Knowledge of techniques for collecting, pressing, mounting, labeling, and preserving plant specimens for herbarium use.



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KU6. Understanding the procedures for cataloging, storing, and maintaining plant specimens within herbarium or museum collections.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Accurate observation of plant morphology.
- GS2.** Skilled in dissection and specimen preservation.
- GS3.** Efficient field data recording and documentation.
- GS4.** Use of field guides and keys for identification.
- GS5.** Correct use of scientific names and terms.
- GS6.** Basic herbarium curation and specimen handling.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Field surveys, collections, preservation and identification of flora</i>	50	50	-	-
PC1. Discuss about the morphological characters of Flowering Plants	10	-	-	-
PC2. Explain the identification Techniques Method of Dissecting Floral Parts and Observation	10	-	-	-
PC3. Explain the identification Techniques of local plant species using field guides and identification keys	10	-	-	-
PC4. Explain to understand and use scientific names	10	-	-	-
PC5. Discuss about Collection of Plants and Field Data; Method of Pressing, Mounting, and Preservation of Plant Specimens	5	-	-	-
PC6. Explain the procedure of Incorporation and Curation of Specimens (Herbarium/ Museum)	5	-	-	-
PC7. Develop a deep understanding of plant morphology and terminology.	-	10	-	-
PC8. Gain hands-on experience in dissecting, collecting, preserving, and curating plant specimens	-	10	-	-
PC9. Enhance skills in identifying plants using various botanical resources and keys.	-	10	-	-
PC10. Build practical experience in field data Collection and specimen documentation.	-	10	-	-
PC11. Understand best practices in plant preservation methods	-	10	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7001
NOS Name	Conduct Field Surveys, Collect, Preserve, and Identify Flora
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N7002: Understand Biodiversity in National and International Contexts

Description

Covers floral and faunal diversity, classification, agro-climatic and bio-geographic zones, wetlands (Amrit Dharohar, Ramsar sites), threats to biodiversity, and sustainable restoration measures.

Scope

The scope covers the following :

- To understand biodiversity, ecological zones, wetlands, and conservation efforts in India and globally.

Elements and Performance Criteria

Biodiversity- An overview (national and international context)

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the overview of floral and faunal diversity in India as well as in the World.
- PC2.** Explain about the general classification scheme and the introduction with floral and faunal groups found in India
- PC3.** Explain about the Concept of different agro-climatic zones as well as bio-geographic zones and their contribution to the biodiversity richness.
- PC4.** Explain about the Concept of Wetlands (Amrit Dharohar) and their types.
- PC5.** Discuss about the Ramsar sites and its faunal diversity in India.
- PC6.** Explain about the various threats to our biodiversity and its impact on our ecosystem.
- PC7.** Explain about the sustainable measures towards the habitat restoration.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic concepts of floral and faunal diversity in India and the world.
- KU2.** Classification of major plant and animal groups in India.
- KU3.** Role of agro-climatic and bio-geographic zones in biodiversity.
- KU4.** Types and importance of wetlands (Amrit Dharohar).
- KU5.** Significance of Ramsar sites and their faunal diversity.
- KU6.** Key threats to biodiversity and sustainable restoration practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Ability to identify and differentiate major floral and faunal species.
- GS2.** Skills to interpret biodiversity distribution across agro-climatic and bio-geographic zones
- GS3.** Capacity to analyze the ecological role of wetlands and Ramsar sites.
- GS4.** Competence in recognizing biodiversity threats and their ecological impacts.
- GS5.** Ability to suggest and apply basic sustainable habitat restoration practices
- GS6.** Communication skills to spread awareness on biodiversity conservation.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Biodiversity- An overview (national and international context)</i>	50	-	-	-
PC1. Describe the overview of floral and faunal diversity in India as well as in the World.	9	-	-	-
PC2. Explain about the general classification scheme and the introduction with floral and faunal groups found in India	7	-	-	-
PC3. Explain about the Concept of different agro-climatic zones as well as bio-geographic zones and their contribution to the biodiversity richness.	5	-	-	-
PC4. Explain about the Concept of Wetlands (Amrit Dharohar) and their types.	10	-	-	-
PC5. Discuss about the Ramsar sites and its faunal diversity in India.	5	-	-	-
PC6. Explain about the various threats to our biodiversity and its impact on our ecosystem.	5	-	-	-
PC7. Explain about the sustainable measures towards the habitat restoration.	9	-	-	-
NOS Total	50	-	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7002
NOS Name	Understand Biodiversity in National and International Contexts
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N7003: Study and Classify Invertebrates and Vertebrates Using Taxonomy

Description

Covers classification, collection, preservation, slide preparation, cataloguing, and field-based identification of zoological specimens, including species of economic, medical, and traditional importance.

Scope

The scope covers the following :

- To provide integrated knowledge and practical skills in animal taxonomy, preservation, anatomical structures, and biodiversity documentation across invertebrates and vertebrates.

Elements and Performance Criteria

Taxonomy: Invertebrate and Vertebrate

To be competent, the user/individual on the job must be able to:

- PC1.** Describe characteristics, collection and preservation techniques of different invertebrates.
- PC2.** Explain key concept of different major and minor invertebrate phylums and general classification scheme (Protista, Mesozoa, Porifera, Cnidaria, Ctenophora, Platyhelminthes, Rotifera, Gastrotricha, Kinorhyncha, Nematoda, Acanthocephala, Sipuncula, Annelida, Onychophora, Arthropoda, Phoronida, Bryozoa, Entoprocta, Brachiopoda, Chaetognatha, Tardigrada, Mollusca, Nemertea, Echinodermata and Hemichordata)
- PC3.** Describe key concept of different lower chordates.
- PC4.** Discuss characteristics and preservation techniques of different vertebrates.
- PC5.** Introduce concept of different taxonomic classes of vertebrates (Pisces, Amphibia, Reptilia, Aves, and Mammals)
- PC6.** Introduce concept of exoskeletal structures like hair/ fur, feathers, horn/antlers.
- PC7.** Explain concept of bones
- PC8.** Introduce cataloguing and Database preparation
- PC9.** Explain maintenance of collections/ museum specimens
- PC10.** Introduce preparation of slides/ pinning technique.
- PC11.** Identify Common species for economic/ medical importance/ traditional knowledge.
- PC12.** Illustrate trichotaxonomy technique.
- PC13.** Visit to Nature trail/ Visit to zoological garden

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic classification of invertebrates, lower chordates, and vertebrates.
- KU2.** Key anatomical features like bones, feathers, hair, and horns.

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- KU3.** Techniques for collection, preservation, slide preparation, and pinning.
- KU4.** Methods for cataloguing and maintaining museum specimens.
- KU5.** Identification of species with economic, medical, or traditional value.
- KU6.** Field skills and basic understanding of trichotaxonomy.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to identify and classify a wide range of invertebrate and vertebrate species.
- GS2.** Proficiency in specimen collection, preservation, slide preparation, and pinning techniques.
- GS3.** Skill in organizing, cataloguing, and managing biological specimen databases.
- GS4.** Capability to maintain and conserve museum and zoological collections.
- GS5.** Field observation and documentation skills through nature trails and zoological visits.
- GS6.** Application of trichotaxonomy and understanding of species with economic or traditional value.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Taxonomy: Invertebrate and Vertebrate</i>	50	50	-	-
PC1. Describe characteristics, collection and preservation techniques of different invertebrates.	8	-	-	-
PC2. Explain key concept of different major and minor invertebrate phylums and general classification scheme (Protista, Mesozoa, Porifera, Cnidaria, Ctenophora, Platyhelminthes, Rotifera, Gastrotricha, Kinorhyncha, Nematoda, Acanthocephala, Sipuncula, Annelida, Onychophora, Arthropoda, Phoronida, Bryozoa, Entoprocta, Brachiopoda, Chaetognatha, Tardigrada, Mollusca, Nemertea, Echinodermata and Hemichordata)	7	-	-	-
PC3. Describe key concept of different lower chordates.	10	-	-	-
PC4. Discuss characteristics and preservation techniques of different vertebrates.	2	-	-	-
PC5. Introduce concept of different taxonomic classes of vertebrates (Pisces, Amphibia, Reptilia, Aves, and Mammals)	6	-	-	-
PC6. Introduce concept of exoskeletal structures like hair/ fur, feathers, horn/antlers.	8	-	-	-
PC7. Explain concept of bones	9	-	-	-
PC8. Introduce cataloguing and Database preparation	-	7	-	-
PC9. Explain maintenance of collections/ museum specimens	-	8	-	-
PC10. Introduce preparation of slides/ pinning technique.	-	8	-	-
PC11. Identify Common species for economic/ medical importance/ traditional knowledge.	-	9	-	-
PC12. Illustrate trichotaxonomy technique.	-	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Visit to Nature trail/ Visit to zoological garden	-	10	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7003
NOS Name	Study and Classify Invertebrates and Vertebrates Using Taxonomy
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N7004: Perform Faunal Surveys, Assess Biodiversity, and Apply Conservation Practices

Description

This NOS covers wildlife survey methods, conservation practices, biodiversity assessment, and legal aspects, with hands-on fieldwork and data analysis.

Scope

The scope covers the following :

- This NOS offers practical knowledge of faunal surveys, conservation strategies, and wildlife laws, with emphasis on fieldwork and ecological analysis.

Elements and Performance Criteria

Faunal survey, assessment, interpretation and conservation practices. Important Acts and Treaties towards Protection of Wildlife in India

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss conceptual idea of different sampling technique and survey/ census methods like transect, quadrat etc.
- PC2.** Introduce protected Areas and corridors
- PC3.** Discuss different biodiversity indices
- PC4.** Introduce in-situ and ex-situ conservation theoretical concept.
- PC5.** Analyze the ways to protect, conserve and preserve biodiversity
- PC6.** Inspect ecological requirements and maintaining factors for priority species and ecosystems
- PC7.** Visit to Nature trail for Survey
- PC8.** Observe behavior of animals in zoological garden
- PC9.** Introduction to data documentation and analysis in spread sheet
- PC10.** Inspect wildlife related judgment in creating new understanding
- PC11.** Build and apply new ideas, methods, and ways of thinking
- PC12.** Introduced to field study procedures
- PC13.** Describe managing Human – Wildlife conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand various wildlife sampling and census methods like transects and quadrats.
- KU2.** Gain knowledge of protected areas, wildlife corridors, and conservation strategies (in-situ & ex-situ).
- KU3.** Learn to assess biodiversity using standard indices and ecological parameters.

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KU4. Recognize habitat needs of priority species and factors influencing their survival.

KU5. Develop skills in data recording, spreadsheet analysis, and field study techniques.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Apply appropriate wildlife sampling and survey techniques in field settings.

GS2. Interpret biodiversity indices and conservation data effectively.

GS3. Demonstrate field observation skills in natural and controlled environments.

GS4. Record, organize, and analyze ecological data using spreadsheets.

GS5. Communicate findings and conservation ideas clearly and logically.

GS6. Solve conservation challenges through critical thinking and innovative approaches.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Faunal survey, assessment, interpretation and conservation practices. Important Acts and Treaties towards Protection of Wildlife in India</i>	32	68	-	-
PC1. Discuss conceptual idea of different sampling technique and survey/ census methods like transect, quadrate etc.	8	-	-	-
PC2. Introduce protected Areas and corridors	6	-	-	-
PC3. Discuss different biodiversity indices	8	-	-	-
PC4. Introduce in-situ and ex-situ conservation theoretical concept.	10	-	-	-
PC5. Analyze the ways to protect, conserve and preserve biodiversity	-	6	-	-
PC6. Inspect ecological requirements and maintaining factors for priority species and ecosystems	-	9	-	-
PC7. Visit to Nature trail for Survey	-	9	-	-
PC8. Observe behavior of animals in zoological garden	-	8	-	-
PC9. Introduction to data documentation and analysis in spread sheet	-	10	-	-
PC10. Inspect wildlife related judgment in creating new understanding	-	4	-	-
PC11. Build and apply new ideas, methods, and ways of thinking	-	5	-	-
PC12. Introduced to field study procedures	-	8	-	-
PC13. Describe managing Human – Wildlife conflict	-	9	-	-
NOS Total	32	68	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7004
NOS Name	Perform Faunal Surveys, Assess Biodiversity, and Apply Conservation Practices
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N7005: Implement the Biological Diversity Act, 2002 and Institutional Mechanisms

Description

Covers key provisions of the Biological Diversity Act & Rules, regulatory frameworks for access and benefit-sharing (ABS), roles of NBA, SBBs/UBCs, and BMCs, registration and levy processes for biological resource users, and hands-on exposure to eco-friendly livelihood techniques like vermicomposting, aquaculture, and apiculture.

Scope

The scope covers the following :

- To equip learners with comprehensive knowledge of biodiversity laws, governance mechanisms, traditional knowledge systems, and sustainable livelihood practices

Elements and Performance Criteria

Biological Diversity Act, 2002 and its Institutional Mechanism for Implementation

To be competent, the user/individual on the job must be able to:

- PC1.** Explain about the Key provisions and objectives of the Biological Diversity Act (2002) & Biological Diversity Rule (2004)
- PC2.** Discuss about the Regulation of Access to Biological Diversity.
- PC3.** Explain about the National Biodiversity Authority; State Biodiversity Boards (SBBs)/ Union Biodiversity Councils (UBCs) and Biodiversity Management Committees (BMC)
- PC4.** Describe about the Registration of traders/ companies who are collecting biological resources
- PC5.** Discuss about the Collection of levy fee from the users of the biological resources
- PC6.** Introduced to various types of livelihood
- PC7.** Introduced to hands-on experience of traditional knowledge for sustainable environment like Vermicomposting technology, Aquaculture, Apiculture, Sericulture, Lac-culture and agricultural methods etc.
- PC8.** Explained about ABS application procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Key provisions and objectives of the Biological Diversity Act, 2002 and Rules, 2004.
- KU2.** Procedures for access and benefit-sharing (ABS) of biological resources.
- KU3.** Roles of NBA, SBBs/UBCs, and BMCs in biodiversity governance.
- KU4.** Registration process for traders/companies using biological resources.
- KU5.** Levy fee collection and benefit distribution to local communities.
- KU6.** Traditional knowledge-based sustainable practices like vermicomposting, aquaculture, apiculture, etc

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to interpret and apply provisions of the Biological Diversity Act and Rules.
- GS2.** Skills to follow procedures for access and benefit-sharing (ABS) applications.
- GS3.** Capacity to coordinate with NBA, SBBs/UBCs, and BMCs for regulatory compliance.
- GS4.** Competence in handling registration and documentation for biological resource users.
- GS5.** Basic understanding of sustainable livelihood practices and traditional eco-technologies.
- GS6.** Communication and awareness skills for engaging communities in biodiversity conservation and benefit-sharing.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Biological Diversity Act, 2002 and its Institutional Mechanism for Implementation</i>	60	40	-	-
PC1. Explain about the Key provisions and objectives of the Biological Diversity Act (2002) & Biological Diversity Rule (2004)	10	-	-	-
PC2. Discuss about the Regulation of Access to Biological Diversity.	10	-	-	-
PC3. Explain about the National Biodiversity Authority; State Biodiversity Boards (SBBs)/ Union Biodiversity Councils (UBCs) and Biodiversity Management Committees (BMC)	10	-	-	-
PC4. Describe about the Registration of traders/ companies who are collecting biological resources	20	-	-	-
PC5. Discuss about the Collection of levy fee from the users of the biological resources	10	-	-	-
PC6. Introduced to various types of livelihood	-	10	-	-
PC7. Introduced to hands-on experience of traditional knowledge for sustainable environment like Vermicomposting technology, Aquaculture, Apiculture, Sericulture, Lac-culture and agricultural methods etc.	-	10	-	-
PC8. Explained about ABS application procedure	-	20	-	-
NOS Total	60	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7005
NOS Name	Implement the Biological Diversity Act, 2002 and Institutional Mechanisms
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N7006: Constitute Biodiversity Management Committees and Operate them Effectively

Description

Covers BMC elections, biodiversity planning, fund management, record-keeping, seed banks, nurseries, and convergence with PRI and GPDP.

Scope

The scope covers the following :

- Strengthen BMCs for biodiversity conservation, sustainable use, and integration with local governance.

Elements and Performance Criteria

Constitution of Biodiversity Management Committee & its operational technique

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss election of Members and establishing administrative setup.
- PC2.** Discuss about the Management of Biodiversity Heritage Sites and native / Indigenous varieties/ breeds.
- PC3.** Explain about the Preparation of Local Biodiversity Action Plan
- PC4.** Explain convergence of BMC with PRI and line departments in particular; Incorporating activities of the BMC in the Gram Panchayat Development Plan
- PC5.** Introduced to maintenance of records and Registers and convening the meeting of the BMCs
- PC6.** Explained financial Matters – opening Bank accounts (Local Biodiversity Fund Account) and its maintenance
- PC7.** Described about the procedure of establishing and managing local seed banks
- PC8.** Establish nursery for medicinal plants and commercial viable plants

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the process of electing BMC members and setting up administrative structures.
- KU2.** Knowledge of managing Biodiversity Heritage Sites and conserving native/indigenous species and breeds.
- KU3.** Familiarity with the steps involved in preparing a Local Biodiversity Action Plan (LBAP).
- KU4.** Understanding of convergence of BMCs with PRI institutions and integration into GPDP.
- KU5.** Ability to maintain BMC records, conduct meetings, and manage Local Biodiversity Funds.
- KU6.** Knowledge of establishing and managing local seed banks and nurseries for medicinal and commercially viable plants.

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to facilitate BMC elections and establish basic administrative systems.
- GS2.** Skills in coordinating biodiversity conservation activities and managing heritage sites.
- GS3.** Competence in preparing and implementing Local Biodiversity Action Plans (LBAP).
- GS4.** Capacity to collaborate with PRI and line departments for integrating BMC activities in GPDP.
- GS5.** Proficiency in maintaining records, conducting BMC meetings, and managing biodiversity funds.
- GS6.** Practical skills in setting up and managing seed banks and nurseries for native and medicinal plants.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Constitution of Biodiversity Management Committee & its operational technique</i>	50	50	-	-
PC1. Discuss election of Members and establishing administrative setup.	10	-	-	-
PC2. Discuss about the Management of Biodiversity Heritage Sites and native / Indigenous varieties/ breeds.	10	-	-	-
PC3. Explain about the Preparation of Local Biodiversity Action Plan	20	-	-	-
PC4. Explain convergence of BMC with PRI and line departments in particular; Incorporating activities of the BMC in the Gram Panchayat Development Plan	10	-	-	-
PC5. Introduced to maintenance of records and Registers and convening the meeting of the BMCs	-	10	-	-
PC6. Explained financial Matters - opening Bank accounts (Local Biodiversity Fund Account) and its maintenance	-	10	-	-
PC7. Described about the procedure of establishing and managing local seed banks	-	10	-	-
PC8. Establish nursery for medicinal plants and commercial viable plants	-	20	-	-
NOS Total	50	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7006
NOS Name	Constitute Biodiversity Management Committees and Operate them Effectively
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N7007: Document People Biodiversity Registers and Develop Bioresource Strategies

Description

The program covers PBR preparation, ecological and socio-economic aspects, bio-resource marketing, and field-based learning for biodiversity

Scope

The scope covers the following :

- To provide knowledge and practical exposure on biodiversity documentation, sustainable use of bio-resources, and related government schemes.

Elements and Performance Criteria

PBR Documentation & Importance, Bio-resource management and Future strategies

To be competent, the user/individual on the job must be able to:

- PC1.** Understand People's Biodiversity Register
- PC2.** Describe about the importance of PBR; PBR documentation Process, Methodology and validation
- PC3.** Explain about the Landscape, Lifescape, Waterscape and Soil types.
- PC4.** Brief overview on Socio-economy
- PC5.** Explain about the sustainable use of Biodiversity through Flagship Programmes and Integrated Watershed Management Programme
- PC6.** Explain about the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)
- PC7.** List of Centrally Sponsored Schemes and its Objectives and activities
- PC8.** Explain about Green Revolution
- PC9.** Understand bio-resources
- PC10.** Discuss marketing of bio-resource-based products in India with example
- PC11.** Field visits and Hands-on experience on PBR preparation of selected locations.
- PC12.** Introduce to Fauna and Flora documentation (wild as well as domesticated including aquatic - both freshwater and marine).
- PC13.** Illustrate Map preparation on documented reports.
- PC14.** Introduce marketing of bio-resources and naturally grown agricultural products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basics and importance of Peoples Biodiversity Register (PBR) and its documentation process.
- KU2.** Key ecological elements: landscape, lifescape, waterscape, and soil types.
- KU3.** Role of socio-economy and government schemes like MGNREGA in biodiversity conservation.

Qualification Pack

KU4. Concepts of bio-resources, flora, fauna, and sustainable use practices.

KU5. Basics of marketing bio-resource products and hands-on PBR fieldwork and mapping.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to document biodiversity using PBR formats with proper methodology and validation.

GS2. Skills to identify and classify landscapes, flora, fauna, soil, and aquatic ecosystems.

GS3. Competence in understanding and linking socio-economic data with biodiversity conservation.

GS4. Familiarity with government schemes and capacity to apply them in sustainable biodiversity management.

GS5. Practical skills in field data collection, mapping, and marketing of bio-resource-based products.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>PBR Documentation & Importance, Bio-resource management and Future strategies</i>	76	24	-	-
PC1. Understand People's Biodiversity Register	8	-	-	-
PC2. Describe about the importance of PBR; PBR documentation Process, Methodology and validation	7	-	-	-
PC3. Explain about the Landscape, Lifescape, Waterscape and Soil types.	8	-	-	-
PC4. Brief overview on Socio-economy	6	-	-	-
PC5. Explain about the sustainable use of Biodiversity through Flagship Programmes and Integrated Watershed Management Programme	7	-	-	-
PC6. Explain about the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)	9	-	-	-
PC7. List of Centrally Sponsored Schemes and its Objectives and activities	11	-	-	-
PC8. Explain about Green Revolution	6	-	-	-
PC9. Understand bio-resources	6	-	-	-
PC10. Discuss marketing of bio-resource-based products in India with example	8	-	-	-
PC11. Field visits and Hands-on experience on PBR preparation of selected locations.	-	4	-	-
PC12. Introduce to Fauna and Flora documentation (wild as well as domesticated including aquatic - both freshwater and marine).	-	9	-	-
PC13. Illustrate Map preparation on documented reports.	-	5	-	-
PC14. Introduce marketing of bio-resources and naturally grown agricultural products	-	6	-	-
NOS Total	76	24	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7007
NOS Name	Document People Biodiversity Registers and Develop Bioresource Strategies
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N7008: Introduce Remote Sensing and GIS Techniques for Conservation Efforts

Description

It provides foundational knowledge in remote sensing and GIS, covering spatial data types, software tools like QGIS, data collection using GPS, map creation, visualization, and analysis techniques for effective spatial decision-making and species distribution mapping.

Scope

The scope covers the following :

- It focuses on the foundational understanding and practical applications of Remote Sensing and GIS technologies, integrating spatial data analysis, visualization, map creation, and field data collection using modern tools and open-source software like QGIS.

Elements and Performance Criteria

Fundamental of Remote Sensing and Hands on training on GIS for conservation

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss about Conceptual background remote sensing technology
- PC2.** Describe vector and raster data as well as grid system.
- PC3.** Explain about GIS data, system operation and interpretation of results.
- PC4.** Understand basic GIS concepts, data types, and applications.
- PC5.** Explain the gain proficiency in using GIS software and tools for spatial data analysis and map creation
- PC6.** Discuss about the methods for collecting, inputting, and managing spatial data.
- PC7.** Develop skills in creating effective maps and visualizations
- PC8.** Introduced to basic training in preparation of map using open-source software like QGIS.
- PC9.** Practice with subset data using selections and feature attributes
- PC10.** Illustrate Data presentation of particular area on the basis of species distribution.
- PC11.** Introduced to Computer based learning technique
- PC12.** Visualize spatial data on a map. Add, remove, and organize data layers.
- PC13.** Introduced to the user interface and basic tools.
- PC14.** Discuss use of GPS for collecting spatial coordinates in the field.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the principles of remote sensing and its application in spatial data acquisition.
- KU2.** Gain knowledge of GIS data formats including vector, raster, and grid systems.
- KU3.** Learn core GIS concepts, system functions, and result interpretation for spatial analysis.

Qualification Pack

- KU4.** Acquire proficiency in using GIS tools (e.g., QGIS) for data visualization and map creation.
- KU5.** Understand methods of spatial data collection, input, management, and GPS-based field data capture
- KU6.** Develop the ability to handle and present spatial data, including species distribution and layer management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to operate GIS and remote sensing software for spatial data analysis and visualization.
- GS2.** Skill in creating, editing, and managing maps using tools like QGIS.
- GS3.** Competence in collecting and processing spatial data using GPS and other input methods.
- GS4.** Capability to interpret and present geographic data effectively, including species distribution.
- GS5.** Proficiency in handling different data formats (vector, raster) and applying attribute-based queries.
- GS6.** Familiarity with computer-based learning tools and user interfaces for spatial data handling.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental of Remote Sensing and Hands on training on GIS for conservation</i>	51	49	-	-
PC1. Discuss about Conceptual background remote sensing technology	10	-	-	-
PC2. Describe vector and raster data as well as grid system.	8	-	-	-
PC3. Explain about GIS data, system operation and interpretation of results.	6	-	-	-
PC4. Understand basic GIS concepts, data types, and applications.	9	-	-	-
PC5. Explain the gain proficiency in using GIS software and tools for spatial data analysis and map creation	6	-	-	-
PC6. Discuss about the methods for collecting, inputting, and managing spatial data.	6	-	-	-
PC7. Develop skills in creating effective maps and visualizations	6	-	-	-
PC8. Introduced to basic training in preparation of map using open-source software like QGIS.	-	5	-	-
PC9. Practice with subset data using selections and feature attributes	-	5	-	-
PC10. Illustrate Data presentation of particular area on the basis of species distribution.	-	4	-	-
PC11. Introduced to Computer based learning technique	-	10	-	-
PC12. Visualize spatial data on a map. Add, remove, and organize data layers.	-	5	-	-
PC13. Introduced to the user interface and basic tools.	-	12	-	-
PC14. Discuss use of GPS for collecting spatial coordinates in the field.	-	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	51	49	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N7008
NOS Name	Introduce Remote Sensing and GIS Techniques for Conservation Efforts
Sector	Green Jobs
Sub-Sector	
Occupation	Conservationist
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0703.Explore Floristic Diversity, Endemism, and Economic Botany for Conservation	53	47	-	-	100	10
SGJ/N7001.Conduct Field Surveys, Collect, Preserve, and Identify Flora	50	50	-	-	100	10
SGJ/N7002.Understand Biodiversity in National and International Contexts	50	-	-	-	50	10
SGJ/N7003.Study and Classify Invertebrates and Vertebrates Using Taxonomy	50	50	-	-	100	10
SGJ/N7004.Perform Faunal Surveys, Assess Biodiversity, and Apply Conservation Practices	32	68	-	-	100	10
SGJ/N7005.Implement the Biological Diversity Act, 2002 and Institutional Mechanisms	60	40	-	-	100	10
SGJ/N7006.Constitute Biodiversity Management Committees and Operate them Effectively	50	50	-	-	100	10
SGJ/N7007.Document People Biodiversity Registers and Develop Bioresource Strategies	76	24	-	-	100	10

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N7008.Introduce Remote Sensing and GIS Techniques for Conservation Efforts	51	49	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	492	408	-	-	900	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.